

Emission Lab Air Specific API Documentation (Versios XX. XX)

Concepts:

This API provides estimations of carbon emissions for flights between specified airports, incorporating various calculation options and environmental impact metrics. It is designed to assist in understanding and mitigating the carbon footprint associated with air travel.

AI Emission Lab Carbon Emission Calculation and Offset Options

Concepts At AI Emission Lab, a "Flight Bundle" is a carefully crafted collection of travel products and services designed by sustainability experts. These bundles aim to mitigate carbon footprint by including eco-conscious options and optimizing travel solutions for minimal environmental impact. Bundles and carbon offset offers are managed through the AI Em,ission Lab Green Portal.

A "Carbon Offset Offer" is a personalized, temporary package created for individual travelers, based on their specific trip details and preferences. These offers include not only travel products but also carbon offset options, calculated based on precise trip parameters. The offset offers are valid from a minimum of 30 minutes to a maximum of 48 hours, ensuring real-time carbon emissions calculations and providing the most effective offset solutions at the time of booking. The expiration time of offset offers is communicated via the API response.

Carbon Offset API Details Carbon offset delivery details are included in the flight bundle confirmation response. AI Emission Lab provides a wide array of carbon offset solutions with distinct delivery methods. Each product will include specifics on how the offset will be executed, including documentation of offset certificates, digital redemption codes for carbon credits, and/or links to offset project pages. For example, if a traveler opts for the "Forestation Project Offset," they will receive information such as:

- **Description:** Contribution towards reforestation in deforested regions.
- **Certificate Codes:** Unique codes confirming participation in the carbon offset initiative.
- **Links:** Access to project dashboards, detailed offset calculations, and comprehensive impact reports.

Through AI Emission Lab, travelers are empowered to make informed decisions and actively participate in reducing their carbon footprint while exploring the world.

API Workflow : [Farabi to Add]

Request parameter:

Emission Calculation:

- **Airport From/To:** Identify the departure and arrival airports.
- **Distance:** The distance in kilometers between the two airports.
- **Flight Class:** Determines emission factors based on the class of flight (e.g., Economy, Business).

Result Structure

Data Elements

- **Airports:**
 - **airport_from:** Departure airport name
 - **airport_to:** Arrival airport name
 - **iata_airport_from:** IATA code for departure airport
 - **iata_airport_to:** IATA code for arrival airport
- **Emissions:**
 - **co2e_gm:** Emissions in grams of CO2 equivalent
 - **co2e_kg:** Emissions in kilograms of CO2 equivalent
 - **co2e_lb:** Emissions in pounds of CO2 equivalent
 - **co2e_mt:** Emissions in metric tons of CO2 equivalent
- **Environmental Impact:**
 - **airQualityImprovement**
 - **carEquivalent**
 - **homeEquivalent**

- **speciesProtected**
- **treesRequired**: Number of trees needed to compensate for emissions
- **waterSaved**: Water saved (liters)
- **Flight Details:**
 - **flight_class**: Average or specified class type
 - **number_of_passengers**: Total number of passengers
 - **round_trip**: Indicates if the trip is round-trip

Metadata

- **Timestamp**: Timestamp of when the calculation was processed.
- **Transaction ID**: Unique identifier for the transaction.
- **Status and Success Flags**: Indicators of API response and operation success.

Disclaimer

The carbon emission calculations are estimates and may vary based on multiple factors such as flight stopovers and layovers, which are not accounted for in this calculation.

Guidelines

Impact Metrics

These metrics provide supplementary information on environmental impact beyond direct carbon emissions:

- **airQualityImprovement**
- **carEquivalent**
- **homeEquivalent**
- **speciesProtected**

Methodology

The emissions calculations are based on the ICAO (International Civil Aviation Organization) methodology, utilizing:

- Distance-based factors
- Aircraft type specifications

- Passenger load factors
- Cargo carried

Authentication and Access

This API requires authentication using access tokens. Ensure these tokens are kept secure and used appropriately for data integrity. Access tokens should be utilized following the Bearer method within request headers for seamless operation.

API Endpoints

Calculate Emissions

- **Endpoint:** /calculate-emissions
- **Method:** POST
- **Description:** Initiates the emissions calculation process using input parameters.

Environment Details

The API can be accessed via different environments (e.g., sandbox for testing and production for live use). Proper environment URLs should be noted for accurate integration.

Support

For issues or questions regarding integration, contact the support team or access online documentation resources available within the partner portal.

This documentation template provides a framework to describe the functionality and integration process of your carbon emission calculation API, emphasizing critical aspects to facilitate user understanding and integration. Adjust sections as needed to align with your project's requirements.

Your interactions may be reviewed by the AI CoE team for the purpose of service improvement. Please chat responsibly.

Error Management :